Software Requirements Specification (SRS)

AgriSense: A Bangla-First AI Mobile Platform for Crop Disease Detection, Crop Advice and a Verified Agri Marketplace

1. Introduction

This document describes what AgriSense must do and how well it must do it. AgriSense is a Bangla-first mobile platform for marginal farmers in Bangladesh. A farmer can photograph a sick leaf and get a diagnosis, ask a question by voice instead of typing, receive soil and weather based crop suggestions, and buy the recommended inputs from verified vendors through a marketplace called Haat_Bazar. The whole app can also be controlled by voice: the farmer can say where to go, ask for the weather, or order a product such as a pesticide, and an AI voice agent carries out the request. A news section called Krishi Barta keeps farmers informed, in Bangla and in simple words, about new diseases, insect pests, viruses and nutrient-deficiency problems. Every function can still be used manually by touch.

1.1 Problem Statement

When a crop shows spots, curling or yellowing, a marginal farmer today has to work out the cause alone. The usual routes are asking a neighbor, describing the symptoms to a local shopkeeper, or waiting for an extension officer (SAAO) who is responsible for many villages. Identifying the disease is done by eye and by memory, so it is often wrong. A wrong guess means the wrong spray, wasted money and a disease that keeps spreading.

The advice that would help exists in DAE, BARC and SRDI publications, but it is written in technical language, is mostly text, and is not available at the time of need. Existing apps such as Plantix, Fosli and Krishoker Janala help in parts, but they expect reading and typing, and none of them leads the farmer from a diagnosis to a verified seller of the right product.

Many farmers also depend on knowledge passed down from earlier generations. They know the diseases that their parents and grandparents saw, but they often do not know about new diseases, new insect pests, new viruses, or the diseases that appear when a crop lacks a particular nutrient. In a discussion with an agri expert, the team learned that such new problems are usually noticed only after a large part of the crop is already damaged, and that there is no regular, trusted and easy-to-understand source that keeps farmers updated.

AI is needed because the manual steps involved can be done faster and more consistently by software: recognizing a disease from a photo (computer vision), understanding a spoken Bangla question (speech recognition), finding the right advice in verified documents (retrieval with an LLM), and choosing suitable crops from soil, district and season data (a trained recommender). AI also collects, sorts and summarizes new disease and pest news into short Bangla updates, which an agri expert checks before the farmers see them. Without these, the app would only be an ordinary shopping app.

1.2 Purpose

The primary objectives of the software are:

- To identify diseases of rice, potato and tomato from a leaf photo and show a confidence score with every result.
- To give treatment advice in Bangla, as text and as voice, using only verified agricultural sources.
- To let farmers ask questions by speaking in Bangla and receive grounded spoken answers.
- To suggest the three most suitable crops for the farmer's district, soil type, season and weather.
- To send low-confidence diagnoses to a human expert so that the farmer is not left with a guess.
- To connect each diagnosis to treatment products sold by admin-verified vendors, and let the farmer order them.
- To let a farmer control the whole app by speaking in Bangla, for example opening the marketplace, hearing the weather update or ordering a recommended pesticide, while keeping every function available by manual touch as well.
- To keep farmers regularly informed, through an auto-updated and expert-verified news section called Krishi Barta, about new diseases, insect pests, viruses and nutrient-deficiency problems of rice, potato and tomato.

1.3 Scope

The system will

- Run as an Android mobile app (Bangla and English) with a web panel for admins and experts.
- Register and authenticate farmers, vendors, experts and admins, with role-based access.
- Detect diseases from leaf photos of rice, potato and tomato (about 12 disease and healthy classes) and show treatment steps.
- Answer Bangla voice questions from a verified knowledge base, and read the answers aloud.
- Let a farmer control the app by voice from any screen: open a feature by saying its name, ask for the weather update, search Haat_Bazar and order a product (for example the pesticide recommended after a diagnosis), with an explicit confirmation before any order is placed.
- Keep every voice function available through normal touch screens as well, so that the farmer can switch between voice and manual control at any time.
- Recommend crops, show weather alerts and a seasonal crop calendar.
- Provide Haat_Bazar: product listing, search, stock update, order placement, status tracking and a digital invoice.
- Store photos taken without internet and upload them when the connection returns.

- Provide Krishi Barta: a regularly auto-updated news section in Bangla, verified by an agri expert, about new and emerging diseases, insect pests, viruses and nutrient-deficiency disorders of rice, potato and tomato, with district-based outbreak alerts, voice reading and offline access to recent items.

The system will not

- Use IoT sensors or drones, or handle crops and insect pests beyond the three selected crops and their listed diseases.
- Process online payments or arrange delivery logistics. Orders in this version use cash on delivery or direct payment between buyer and seller.
- Run AI models on the phone itself, or release an iOS app or Play Store version. These are future work.
- Provide voice control for vendors, experts or admins in this version. They use the screens and the web panel manually. Voice control covers Bangla and English only.
- Publish general agricultural, market or price news in Krishi Barta. In this version it covers only the three selected crops, and every item is published only after an agri expert approves it.

AI boundaries

The AI will classify leaf photos, report how confident it is, convert Bangla speech to text and back, retrieve and summarize advice from verified documents, and rank crops. The AI will not give a final or guaranteed diagnosis, will not suggest pesticide doses that are absent from the verified knowledge base, will not answer questions outside agriculture, and will not replace agricultural experts. The Krishi Barta news engine only collects, sorts and summarizes items from approved sources. It will not publish anything before an agri expert approves it, will not add treatment or pesticide doses that are not in the source or the verified knowledge base, and will not present a rumor or an unverified report as news. The voice command agent will only map speech to a fixed list of app actions (open a screen, give the weather, search a product, order a product). It will not carry out any action outside that list, will not place or cancel an order without the farmer's explicit confirmation, and will ask the farmer to repeat or choose when it is not sure what was meant.

2. Design and Implementation Constraints

2.1 Programming Language

Part	Technology	Why we chose it
Mobile app	Dart with Flutter	One codebase, good camera and microphone support, and it is easy to build large-button, high-contrast screens for farmers.
Backend	Python with Django and Django REST Framework	Built-in authentication and admin panel, and the same language as our AI code, so integration is simpler.
AI code	Python	TensorFlow/Keras, scikit-learn and OpenCV are all available for Python.

Part	Technology	Why we chose it
Queries	SQL	Used through the Django ORM for the relational data.

2.2 Database & Servers

- PostgreSQL stores users, diagnoses, products, orders, alerts and feedback, because the data is relational and needs consistency, especially for orders and stock.
- Leaf photos are stored in cloud object storage, and the database stores only the image URL.
- A vector database (ChromaDB or pgvector) stores embeddings of the verified knowledge base documents for retrieval.
- The mobile app talks to the Django backend through RESTful APIs over HTTPS with JWT authentication.
- The backend and the AI service are hosted on a cloud platform (AWS or GCP student credits). A weather API (such as OpenWeatherMap) supplies forecasts.
- Each voice command, its recognized intent, confidence and outcome are logged in the database for audit and improvement. Raw voice recordings are not kept unless the farmer gives consent.
- Krishi Barta items (title, Bangla summary, category, crop, severity, affected districts, source and status) and their delivery to farmers are stored in PostgreSQL. A scheduled background job (for example a Celery task) collects news from the approved sources every day. Published items are also added to the vector index so that the voice assistant can use them.

2.3 AI Model & Services

AI feature	Approach and service	Justification
Disease detection	MobileNetV2 (transfer learning) built with TensorFlow/Keras, served through a Python FastAPI endpoint called by the Django backend.	Our image dataset is small, so transfer learning works better than training from scratch. The model is light and fast, and can be moved to the phone later.
Voice assistant	Cloud Bangla speech-to-text and text-to-speech, plus RAG: multilingual embeddings, vector search, and an LLM API that writes the answer from the retrieved passages.	Training our own Bangla speech model is not realistic. RAG keeps answers tied to verified documents, which reduces invented advice.
Voice command agent	Cloud Bangla speech-to-text and text-to-speech, plus an LLM API with function calling. The LLM maps the transcript to one action from a fixed list (navigate, weather, search product, order product) and its parameters. The Django backend then runs the action through the existing APIs.	A fixed action list keeps the agent predictable and safe. The same APIs are used by the touch screens, so voice and manual control always give the same result. A confirmation step prevents accidental orders.

AI feature	Approach and service	Justification
Crop recommender	Random forest (scikit-learn) on district-level soil and crop suitability data, combined with simple expert rules.	The data is small and structured, so a tree-based model trains quickly, can be explained, and is easy to check with an expert.
Product matching	Embedding similarity between treatment text and vendor product names and descriptions.	Vendors write product names in different ways, so exact text search would miss matches.
Krishi Barta news engine	Scheduled collection from approved sources, text cleaning, duplicate removal with multilingual embeddings, and an LLM API that tags each item (category, crop, district, severity) and writes a short Bangla summary from the source text only. An agri expert approves every item before it is published.	Official bulletins are long and technical, and farmers need short, simple Bangla. The LLM does the reading and writing work, and the expert check keeps the content correct and prevents invented advice.

3. User Classes and Characteristics

User class	Characteristics	Permissions
Farmer	Main user. Often low literacy and limited smartphone experience. Works outdoors and prefers voice and icons.	Register and manage own profile. Capture or upload photos, view own diagnosis history, ask voice questions, get crop recommendations and weather alerts, read Krishi Barta news and receive outbreak alerts, browse products, place and track own orders, request expert review, give feedback. Control the whole app by voice (open any feature, ask for the weather, search and order products) or manually, with the same permissions in both modes.
Vendor	Licensed seller of seeds, fertilizers and medicines. Moderate smartphone use.	Manage own profile, add, edit and delete own products, update stock, view and update status of orders for own products. Can list products only after admin verification.
Agri Expert	SAAO or agricultural specialist who understands crop diseases.	View the queue of low-confidence diagnoses, confirm or correct them, add and update knowledge base entries, and review, edit, approve or reject Krishi Barta news drafts before publication. Cannot manage users or orders.
System Admin	Technical staff of the project.	Verify, suspend or remove vendors. Manage all user accounts, view system logs, manage weather and soil data, manage the approved news sources and the collection schedule, and monitor system health. Does not edit a diagnosis.

4. Requirement Elicitation

This section explains how the requirements were collected from real users and how they changed. Items highlighted in yellow must be replaced with the team's actual data before submission.

4.1 Elicitation Techniques

Technique	Who took part	When / where	Number
Interviews	Farmers growing like: rice, potato or tomato	9/10/2026 and Bordail, Dhamrai	01826415605
Interview	SAAO or agricultural expert	11/10/2026 and google meet	0 1303-726651
Interview	Local fertilizer / pesticide vendors	9/10/2026 and Savar Rupnagar	01782543710
Questionnaire	Farmers and vendors	9/10/2026 and Savar Nama Bazar	Multiple
Observation	How farmers currently identify disease and buy inputs; review of Plantix, Fosli and Krishoker Janala	9/10/2026	N/A

4.2 Elicitation Evidence

The evidence below is attached in Appendix A:

- Interview notes (and recordings, with consent): File
- Photos of the field visit or market visit: Pictures
- Confirmation or permission appreciation from the expert or office: expert's appreciation

4.3 Key Findings

After eight to ten rejection only few of then want to talk with us. We collect the proper evidence of them. In an addition, they give some genuine ideas and stuff that we implement in our project.

- Finding 1: Consultation 01, Consultation 02, Consultation 03
- Finding 2: the SAAO or expert said
- Finding 3: what vendors said
- Finding 4: In a discussion with the agri expert (Ahosan Habib Simanto, Agricultural Consultant & Researcher, BAU, 11/10/2026), the team learned that most farmers rely on disease knowledge passed down from earlier generations and are not aware of new diseases, insect pests, viruses or nutrient-deficiency disorders. The expert advised that farmers should receive regular, verified updates in simple Bangla. This led to the Krishi Barta feature.

4.4 Requirement Changes After Elicitation

The first draft of the requirements was compared with what users told us. The Before and After columns below show the requirement text, and the last column must state the user input that caused each change.

ID	Change	Before Elicitation	After Elicitation	Source / Reason
FR-6	Modified	The system shall detect the disease from a leaf photo.	The system shall detect the disease from a leaf photo and show a confidence score; below 0.70 it asks for a retake or sends the case to an expert.	Source
FR-10	New	None.	The system shall accept Bangla voice questions and answer from the verified knowledge base.	Source
FR-19	Modified	The system shall support online payment for orders.	Orders use cash on delivery or direct payment between buyer and seller.	Source
FR-27	New	None.	Photos captured offline are queued and uploaded when internet returns.	Source
FR-29 to FR-31, FR-40	New	None.	The system shall let the farmer control the app by voice from any screen: speech is converted to text, the intent is recognized and the requested feature is opened.	Source
FR-34	New	None.	The system shall let the farmer order a product by voice, including the pesticide recommended after a diagnosis, and shall place the order only after confirmation.	Source
FR-31, FR-40	Modified	The voice-controllable features were diagnosis, voice assistant, crop advice, weather, Haat_Bazar, orders, history and profile.	Krishi Barta is added to the screens that the farmer can open, listen to and control by voice.	Added together with Krishi Barta so that low-literacy farmers can open and hear the news by voice (agri expert discussion, see FR-41 to FR-50).
FR-41 to FR-50	New	None.	The system shall provide Krishi Barta, an auto-updated news section in Bangla where farmers receive expert-verified information on new diseases, insect pests, viruses and nutrient-deficiency disorders, with district-based alerts, voice reading and offline access.	Discussion with the agri expert (Ahosan Habib Simanto) farmers mostly rely on knowledge inherited from earlier generations and do not know about new diseases, pests, viruses or nutrient-related disorders. The expert advised a regular, verified update channel.

5. Functional Requirements (FR)

Requirements that use AI are marked (AI).

ID	Requirement Name	Description	Users	Priority
FR-1	User Registration	The system must let farmers, vendors and experts register with name, mobile number, role and district.	Farmer, Vendor, Expert	High
FR-2	Secure Login	The system must authenticate users and send them to the dashboard for their role.	All	High
FR-3	Password Reset	The system must let a user reset the password through an SMS OTP.	Farmer, Vendor, Expert	Medium
FR-4	Profile Management	Users must be able to update their profile, farm location and crop preferences.	Farmer, Vendor	Medium
FR-5	Photo Capture / Upload	The system must let a farmer capture a leaf photo with the in-app camera or upload one from the gallery.	Farmer	High
FR-6	AI Disease Detection (AI)	For an uploaded photo of a rice, potato or tomato leaf, the system must return the most likely disease class and a confidence score between 0 and 1 within 5 seconds.	Farmer	High
FR-7	Treatment Advice (AI)	After a diagnosis, the system must show treatment steps in Bangla taken from the verified knowledge base.	Farmer	High
FR-8	Low-Confidence Handling (AI)	If the confidence score is below 0.70, the system must ask the farmer to retake the photo or send the case to the expert review queue, and must not show it as a final result.	Farmer, Expert	High
FR-9	Diagnosis History	The system must save every diagnosis and let the farmer view the history.	Farmer	Medium
FR-10	Voice Query Input (AI)	The system must accept a Bangla voice question and convert it to text.	Farmer	High
FR-11	Grounded Answer (AI)	The system must answer a question using passages retrieved from the verified knowledge base, and must say that it has no answer when no relevant passage is found.	Farmer	High
FR-12	Bangla Voice Output (AI)	The system must read answers and treatment advice aloud in Bangla.	Farmer	Medium
FR-13	Crop Recommendation (AI)	Given district, soil type, season and current weather, the system must suggest the top three suitable crops, each with a short reason.	Farmer	High
FR-14	Weather Alerts	The system must fetch weather data for the farmer's district and send alerts for heavy rain, heat or similar risks.	Farmer	Medium
FR-15	Seasonal Calendar	The system must show a planting and harvesting calendar for the selected crop.	Farmer	Low
FR-16	Product Matching (AI)	After a diagnosis, the system must list matching treatment products from verified vendors with price and stock.	Farmer	Medium

ID	Requirement Name	Description	Users	Priority
FR-17	Product Search and Filter	The system must let farmers search products by name and category and filter by price and availability.	Farmer	Medium
FR-18	Vendor Product Management	Vendors must be able to add, edit and delete their products and update stock levels.	Vendor	High
FR-19	Order Placement	Farmers must be able to place an order for available products, and the stock must reduce after the vendor confirms.	Farmer, Vendor	High
FR-20	Order Tracking	The system must show order status (Pending, Shipped, Delivered) to the farmer and let the vendor update it.	Farmer, Vendor	Medium
FR-21	Digital Invoice	The system must generate a digital invoice for each order.	Farmer, Vendor	Low
FR-22	Vendor Verification	Admin must be able to verify, reject or suspend vendor accounts, and only verified vendors may list products.	Admin	High
FR-23	User Management	Admin must be able to view, suspend or remove user accounts.	Admin	Medium
FR-24	Expert Review	Experts must be able to view flagged diagnoses, confirm or correct the disease, and send the result back to the farmer.	Expert	High
FR-25	Knowledge Base Update	Experts must be able to add and edit disease and treatment entries, and the retrieval index must be refreshed after a change.	Expert, Admin	Medium
FR-26	Feedback	Users must be able to rate advice and vendors and leave comments.	Farmer, Vendor	Low
FR-27	Offline Photo Queue	Photos captured without internet must be stored on the phone and uploaded automatically when the connection returns.	Farmer	Medium
FR-28	Language Selection	The system must let users switch between Bangla and English, with Bangla as the default.	All	Medium
FR-29	Voice Command Input (AI)	The system must provide a microphone button on every farmer screen, accept a spoken command in Bangla or English and convert it to text.	Farmer	High
FR-30	Intent Recognition (AI)	The system must interpret the command text as one action from a fixed list (open screen, get weather, search product, order product) with its parameters and a confidence score. If the confidence is low, it must ask the farmer to repeat or choose, and must not act.	Farmer	High
FR-31	Voice Navigation	The system must open the feature the farmer names from any screen, for example Diagnosis camera, Voice assistant, Crop advice, Weather, Krishi Barta, Haat_Bazar, Orders, History or Profile. Example: "go to the marketplace".	Farmer	High

ID	Requirement Name	Description	Users	Priority
FR-32	Voice Weather Update	On a spoken request, the system must fetch the current weather, forecast and active alerts for the farmer's district, read them aloud in Bangla and show them on screen.	Farmer	High
FR-33	Voice Product Search	The system must search Haat_Bazar by a spoken product name or category (for example a pesticide) and read out the best results with price and stock.	Farmer	Medium
FR-34	Voice Order Placement	The system must let the farmer order a product by voice. It must read back the product, vendor, quantity, price and payment mode, and must create the order only after the farmer confirms by voice or by tapping. Stock and order status then follow FR-19 and FR-20.	Farmer, Vendor	High
FR-35	Pesticide Order Shortcut (AI)	After a diagnosis, the system must let the farmer say, for example, "order this pesticide". It must select the best matching in-stock product from verified vendors (FR-16) and start the voice order flow.	Farmer	High
FR-36	Voice and Manual Control	Every function that can be done by voice must also be possible by touch. The farmer must be able to switch between the two at any time, turn voice control on or off, and cancel a voice action by tapping.	Farmer	High
FR-37	Voice Feedback	The system must tell the farmer, by voice and on screen, what it understood and what it is doing. If it did not understand, it must ask the farmer to repeat and offer the manual option.	Farmer	Medium
FR-38	Voice Safety Controls	Actions that cost money or change data (placing or cancelling an order) must need explicit confirmation. The agent must refuse requests outside the app and agriculture, and must act only within the logged-in farmer's own account and permissions.	Farmer	High
FR-39	Voice Command Log	The system must record each voice command with its text, recognized intent, confidence and outcome. Voice recordings must not be stored unless the farmer consents.	Farmer, Admin	Low
FR-40	Full-App Voice Control (AI)	The whole farmer app must be controllable by voice: every farmer feature (disease diagnosis, voice assistant, crop recommendation, weather, Krishi Barta news, seasonal calendar, Haat_Bazar search, ordering and tracking, history, profile and language) must be startable and completable by spoken commands, with the same result as manual use. Where the farmer says to go, the app goes there.	Farmer	High

ID	Requirement Name	Description	Users	Priority
FR-41	Krishi Barta News Section (AI)	The system must provide a Krishi Barta section that shows regularly updated news in Bangla about new and emerging diseases, insect pests, viruses and nutrient-deficiency disorders of rice, potato and tomato. Each item must show a title, a short summary, the symptoms, the affected crop, the date, the source and a severity label.	Farmer	High
FR-42	Automated News Collection (AI)	The system must run a scheduled job at least once every 24 hours that collects new items from the approved sources (for example DAE, BARC, BRRI, BARI and SRDI), cleans the text and removes items that duplicate ones already collected or published.	Admin, Expert	High
FR-43	News Classification and Summary (AI)	For each collected item, the system must identify the category (disease, pest, virus or nutrient deficiency), the crop, the affected districts and the severity, and must write a short Bangla summary using only the text of the source item. If the AI is not confident, the item must be flagged for the expert.	Expert	High
FR-44	Expert Verification and Publishing	An agri expert must be able to review each draft, edit it, and approve or reject it with a reason, and must be able to add an item by hand and mark an item as urgent. Only approved items may be published to farmers.	Expert	High
FR-45	News Filter and Personalization	The system must show first the items that match the farmer's crops and district, and must let the farmer filter news by crop, category (disease, pest, virus, nutrient deficiency) and date.	Farmer	Medium
FR-46	Outbreak Alert Notification	When an urgent or severe item matches a farmer's crop and district, the system must send a push notification in Bangla. The farmer must be able to turn notifications on or off.	Farmer	Medium
FR-47	Voice Access to Krishi Barta (AI)	The farmer must be able to open Krishi Barta by voice and have the latest items read aloud in Bangla, with pictures or icons of the symptoms where available, so that low-literacy farmers can use it.	Farmer	Medium
FR-48	Offline News Access	The app must keep the latest 20 verified items on the phone so that they can be read without internet, and must show the date of the last update.	Farmer	Medium
FR-49	Link to Diagnosis and Products	From a Krishi Barta item the farmer must be able to open the diagnosis camera ("Check my crop") and, for a disease or pest item, see matching treatment products from verified vendors (FR-16). Published items must also be added to the knowledge base index so that the voice assistant can answer questions about them.	Farmer	Medium

ID	Requirement Name	Description	Users	Priority
FR-50	News Source Management	Admin must be able to add, disable or remove approved news sources and change the collection schedule. Experts must be able to suggest a new source. The system must record the source of every published item.	Admin, Expert	Medium

6. Non-Functional Requirements

- **Performance:** On a normal 4G connection, the disease result must appear within 5 seconds of upload, and ordinary screens must load within 2 seconds.
- **AI Quality:** The disease classifier must reach at least 85% accuracy on the held-out test set, and recall for each disease class must be reported. If the AI service is unavailable or confidence is low, the system must fall back to the expert review queue or a clear message, and never show an invented result.
- **Security:** Passwords must be stored hashed, all communication must use HTTPS, and users must only reach the data allowed for their role.
- **Privacy:** The app must ask for consent before collecting photos, location and phone numbers, and must collect only what it needs. Voice recordings must be used only to process the command and must be deleted afterwards unless the farmer consents to keep them for improving the system.
- **Usability:** The interface must be icon-based with large buttons, and a farmer must be able to get a diagnosis in no more than four taps after opening the app. Every function available by voice must also be available by touch.
- **Accessibility:** The app must offer a high-contrast mode for outdoor use and voice input and output for low-literacy users.
- **Voice Control Performance:** On a normal 4G connection, a spoken command must lead to the action or a spoken reply within 4 seconds, and the voice command agent must identify the correct intent for at least 90% of the commands in the test set.
- **Voice Control Safety and Fallback:** No order may be created without explicit confirmation. If speech recognition, the agent or the network fails, the app must say so and let the farmer continue with the touch screens, and must never carry out a guessed action.
- **Krishi Barta Freshness and Accuracy:** New items must be collected at least once every 24 hours, and an urgent outbreak notice must reach matching farmers within 1 hour of expert approval. No item may be published without expert approval, and every item must show its source and date. On the expert-labelled test set, the category and crop tags must be correct for at least 90% of the items.
- **Localization:** All screens and advice must be available in Bangla.
- **Reliability:** The system should stay available during the pilot test, and errors must be logged so that they can be fixed.

- **Compatibility:** The app must run on Android 8.0 and above and work on slow networks.
- **Data Integrity:** Data collected offline must sync without being lost or duplicated.
- **Maintainability:** The code must be modular, kept under Git version control and documented.

7. System Design Diagrams

7.1 Use Case Diagram

The diagram shows the four human actors, the AI News Service (the scheduled job that collects news for Krishi Barta) and the main use cases. Orange ovals are the features that depend on AI. They are the voice control use cases (Control Whole App by Voice and Order Product by Voice) and the Krishi Barta use cases (View Krishi Barta, Get Outbreak Alert, Auto-Collect and Summarize News, and Verify and Publish Krishi Barta). Control Whole App by Voice extends every farmer use case (dashed orange group in the diagram), so each of them can be done by voice or manually.

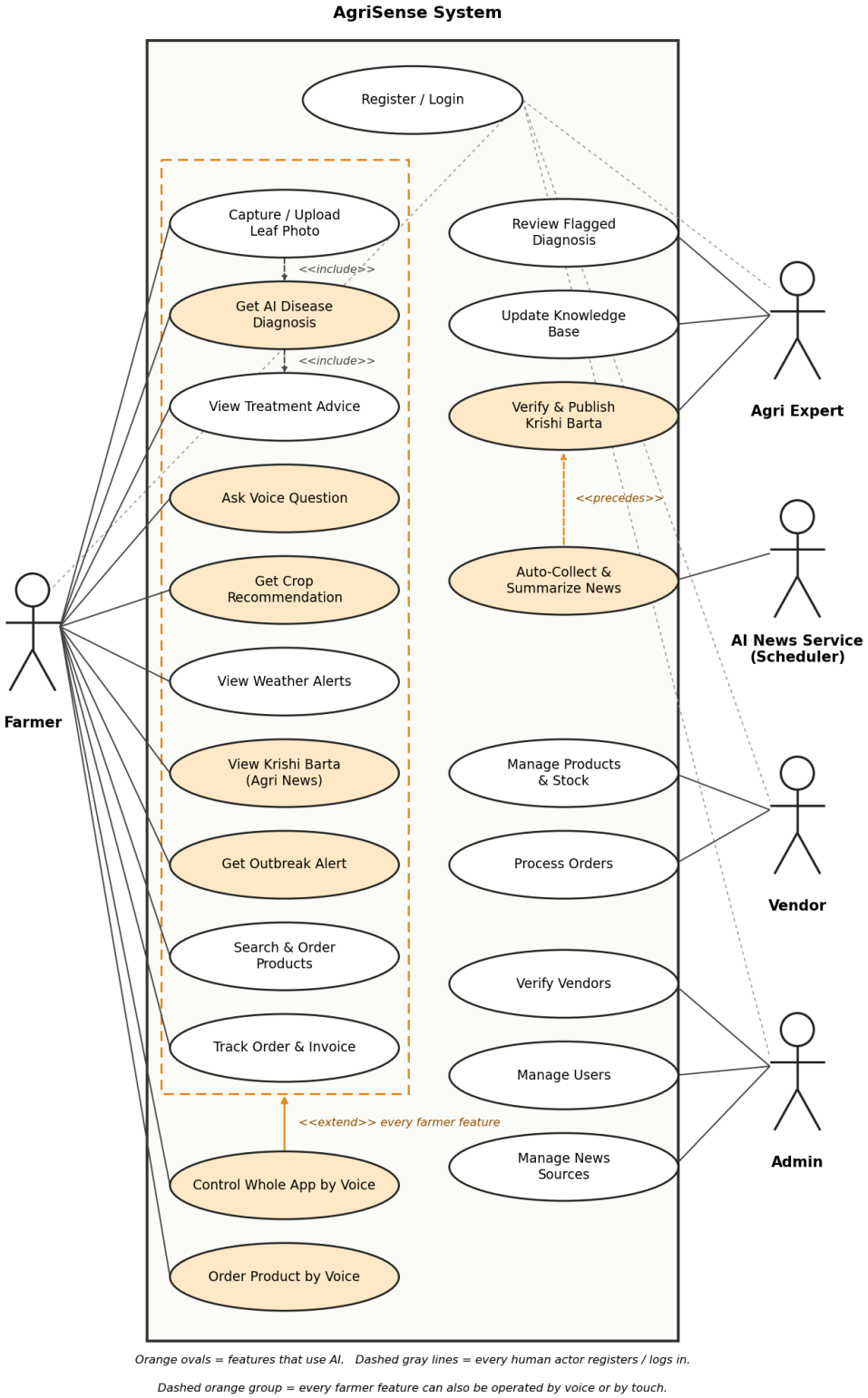


Figure 1: Use case diagram of AgriSense

7.2 Use Case Descriptions

Main success flows are given below for the High priority features.

UC-01	**Detect Crop Disease**
Actor	Farmer (AI service as supporting system)
Related FR	FR-5, FR-6, FR-7, FR-8
Precondition	Farmer is logged in and has a leaf photo of a supported crop.
Main Success Flow	1. Farmer opens the camera screen and captures a photo, or picks one from the gallery. 2. The app uploads the photo to the server. 3. The server preprocesses the image and sends it to the disease model. 4. The model returns a disease class and a confidence score. 5. The score is 0.70 or higher, so the system shows the disease name and Bangla treatment steps from the knowledge base. 6. The system saves the result in the diagnosis history and shows matching products.
Alternate / Exception	No internet at step 2: the photo is queued and uploaded later (FR-27). Confidence below 0.70 at step 5: the system asks for a retake or sends the case to an expert (see UC-04). AI service unavailable: the system sends the case to the expert queue and tells the farmer.
Postcondition	A diagnosis record exists in history and the farmer can continue to the marketplace.

UC-02	**Ask a Voice Question**
Actor	Farmer
Related FR	FR-10, FR-11, FR-12
Precondition	Farmer is logged in, the microphone permission is given and internet is available.
Main Success Flow	1. Farmer taps the microphone and speaks a question in Bangla. 2. The system converts the speech to text. 3. The system searches the knowledge base and retrieves the most relevant passages. 4. The LLM writes a short Bangla answer using only those passages. 5. The system shows the answer as text and reads it aloud.
Alternate / Exception	Speech not understood: the system asks the farmer to repeat or type the question. No relevant passage found: the system says it has no answer and offers to send the question to an expert.
Postcondition	The question and answer are saved, and the farmer has received advice grounded in verified sources.

UC-03	Get Crop Recommendation
Actor	Farmer
Related FR	FR-13, FR-14
Precondition	Farmer is logged in and has set a district.
Main Success Flow	1. Farmer opens the crop advice screen and selects soil type and season. 2. The system fetches the weather forecast for the district. 3. The recommender ranks suitable crops from district, soil, season and weather. 4. The system shows the top three crops, each with a short reason.
Alternate / Exception	Weather API unavailable: the system uses district and season data only and says that weather was not included. No matching data for the district: the system shows general seasonal advice.
Postcondition	The recommendation is displayed and saved for later reference.

UC-04	Review Flagged Diagnosis
Actor	Agri Expert
Related FR	FR-8, FR-24
Precondition	Expert is logged in and at least one low-confidence diagnosis is in the queue.
Main Success Flow	1. Expert opens the review queue and selects a case. 2. The system shows the photo, the AI result and its confidence score. 3. Expert confirms the disease or selects a different one and adds a note. 4. The system updates the diagnosis and the treatment advice. 5. The farmer is notified that the reviewed result is ready.
Alternate / Exception	Expert is unsure: the case is marked as needing a retake and the farmer is asked for clearer photos.
Postcondition	The diagnosis is marked as expert reviewed and can be used to improve the dataset.

UC-05	Place an Order
Actor	Farmer, Vendor
Related FR	FR-17, FR-19, FR-20
Precondition	Farmer is logged in and the product is listed by a verified vendor with stock available.
Main Success Flow	1. Farmer searches or opens a recommended product. 2. Farmer selects the quantity and places the order. 3. The system creates the order with status Pending and notifies the vendor. 4. Vendor confirms the order and the stock is reduced. 5. Vendor updates the status to Shipped and then Delivered. 6. The system generates the digital invoice.
Alternate / Exception	Requested quantity exceeds stock: the system shows the available amount. Vendor rejects the order: the farmer is notified and the stock is unchanged.
Postcondition	The order is recorded with its final status and an invoice.

UC-06	Manage Products and Stock
Actor	Vendor
Related FR	FR-18
Precondition	Vendor is logged in and verified by the admin.
Main Success Flow	1. Vendor opens the product screen and chooses to add a product. 2. Vendor enters name, category, price and stock. 3. The system validates the data and saves the product. 4. The product appears in the marketplace and in matching results.
Alternate / Exception	Vendor not yet verified: the system blocks listing and shows the verification status.
Postcondition	The product is listed and its stock can be updated at any time.

UC-07	Verify Vendor
Actor	System Admin
Related FR	FR-22
Precondition	Admin is logged in and a vendor has registered.
Main Success Flow	1. Admin opens the list of pending vendors. 2. Admin checks the shop name and license details. 3. Admin approves the vendor. 4. The vendor's status changes to Verified and the vendor is notified.
Alternate / Exception	Details are incomplete or doubtful: Admin rejects or suspends the vendor with a reason.
Postcondition	Only verified vendors can list products.

UC-08	Control the Whole App by Voice
Actor	Farmer (voice command agent as supporting system)
Related FR	FR-29, FR-30, FR-31, FR-32, FR-36, FR-37, FR-38, FR-40
Precondition	Farmer is logged in, the microphone permission is given and internet is available.
Main Success Flow	1. Farmer taps the microphone button, which is available on every screen, and speaks a command in Bangla, for example "go to the marketplace" or "what is the weather today". 2. The system converts the speech to text. 3. The voice command agent identifies the intent and its parameters from the fixed list of app actions. 4. The system tells the farmer what it understood, for example "Opening Haat_Bazar". 5. The system performs the action: it opens the requested screen, or fetches the forecast and alerts for the farmer's district and reads them aloud while showing them on screen. 6. The farmer continues by voice, or touches the screen at any time to continue manually.
Alternate / Exception	Speech not understood or confidence low: the system asks the farmer to repeat or shows the options on screen. Request outside the action list: the system says it cannot do that and shows the main menu. Weather API unavailable: the system says the weather cannot be fetched now. No internet or voice service down: the system says so and the farmer uses the touch screens.
Postcondition	The requested screen or information is delivered and the command is logged. The farmer can always continue manually.

UC-09	Order a Product by Voice
Actor	Farmer, Vendor
Related FR	FR-16, FR-19, FR-33, FR-34, FR-35, FR-36, FR-38
Precondition	Farmer is logged in, the microphone permission is given, internet is available, and a verified vendor has the product in stock.
Main Success Flow	1. After a diagnosis, or from any screen, the farmer says, for example, "order this pesticide" or "order two bottles of [product name]". 2. The system converts the speech to text and the agent identifies the order intent, product and quantity. 3. For "this pesticide", the system takes the treatment from the latest diagnosis and selects the best matching in-stock product from verified vendors. For a named product, it searches Haat_Bazar. 4. The system reads out and shows a summary: product, vendor, quantity, price and payment mode (cash on delivery or direct payment). 5. The farmer confirms by saying "yes, confirm" or by tapping Confirm. 6. The system creates the order with status Pending, notifies the vendor and continues as in UC-05 (confirmation, stock update, shipping, invoice). 7. The system tells the farmer that the order is placed and how to track it.
Alternate / Exception	Several products match: the system reads the top options and the farmer chooses by voice or by tapping. Quantity not stated: the system asks for it. Quantity exceeds stock: the system tells the farmer the available amount. Farmer says no, cancels or does not answer: no order is created. No recent diagnosis for "this pesticide": the system asks the farmer to name the product or opens the diagnosis screen. No matching product in stock: the system tells the farmer and offers a manual search.
Postcondition	An order exists only if the farmer confirmed it, with an invoice later as in UC-05. The command and its outcome are logged.

UC-10	View Krishi Barta News
Actor	Farmer (AI News Service and Agri Expert as supporting parties)
Related FR	FR-41, FR-45, FR-46, FR-47, FR-48, FR-49
Precondition	Farmer is logged in and at least one verified item has been published.
Main Success Flow	1. The farmer opens Krishi Barta by touch, or by saying "Krishi Barta" (see UC-08). 2. The app requests the latest published items from the server. 3. The system lists the items, showing first those that match the farmer's crops and district. Each item shows a title, category icon, severity, date and source. 4. The farmer filters by crop or category if needed and opens an item. 5. The system shows the Bangla summary, the symptoms, the affected crop and districts and the advised action, and reads them aloud when the farmer taps the speaker button or voice mode is on. 6. The farmer taps "Check my crop" to open the diagnosis camera (UC-01), or views matching treatment products from verified vendors. 7. The system records that the item was delivered and read.
Alternate / Exception	No internet at step 2: the app shows the latest 20 cached items with the date of the last update (FR-48). No item matches the farmer's crops or district: the system shows the latest items of all three crops. Urgent item while the app is closed: the farmer receives a push notification (FR-46), and tapping it opens the item at step 5. Voice command not understood: the system asks the farmer to repeat or shows the Krishi Barta button.
Postcondition	The farmer has read or heard a verified update, and the delivery and read time are saved.

UC-11	Collect, Verify and Publish Krishi Barta News
Actor	Agri Expert (AI News Service as supporting system; Admin manages the sources)
Related FR	FR-42, FR-43, FR-44, FR-46, FR-49, FR-50
Precondition	The admin has set up the approved sources, the scheduled job is running and the expert is logged in to the web panel.
Main Success Flow	1. The scheduled job starts (at least once every 24 hours) and fetches new items from the approved sources. 2. The system cleans the text and removes items that duplicate earlier ones. 3. The AI tags each new item (category, crop, affected districts, severity) and writes a short Bangla summary using only the source text. 4. The system saves each item as Draft and adds it to the expert's review queue. 5. The expert opens a draft, compares it with the source and edits the summary or tags if needed. 6. The expert approves the item, and marks it urgent if it is a severe outbreak. 7. The system publishes the item, adds it to the knowledge base index and sends push notifications to the farmers whose crop and district match.
Alternate / Exception	A source cannot be reached at step 1: the job retries later and the admin sees the error in the system log. AI confidence is low at step 3, or the LLM is unavailable: the item is marked "needs expert writing" and the expert writes the summary by hand. The expert rejects the item at step 6: it is saved as Rejected with a reason and is never shown to farmers. The expert adds an item by hand: it is saved as Draft with the source "Expert" and continues at step 6.
Postcondition	Only expert-approved items are visible to farmers, and every published item shows its source, date and verifying expert.

UC-12	Register and Log In
Actor	Farmer, Vendor, Agri Expert (the Admin uses only the login steps)
Related FR	FR-1, FR-2, FR-3
Precondition	The app is installed and the user has a mobile number.
Main Success Flow	1. The user opens the app, which is in Bangla by default, and chooses Register or Log in. 2. For registration, the user enters name, mobile number, role, district and a password. 3. The system validates the data, checks that the mobile number is not already registered and creates the account. A vendor account stays unverified until the admin approves it (UC-07). 4. The user logs in with the mobile number and password. 5. The system authenticates the user and opens the dashboard for the user's role.
Alternate / Exception	Mobile number already registered: the system tells the user to log in instead. Wrong mobile number or password: the system shows a message and lets the user retry or reset the password with an SMS OTP (FR-3).
Postcondition	The user is logged in and sees the dashboard allowed for the role.

7.3 Activity Diagram

The activity diagrams below show the flow of complex features. Figure 2 follows a farmer's photo through AI diagnosis and expert review when needed, to ordering the recommended products. Every step in it can be run by voice or by touch. Figure 3 shows how a voice command controls the whole app: the farmer chooses voice or touch, speaks a command, the agent identifies the action, a costly action such as an order is read back and confirmed, and the app runs it through its normal services. Figure 4 shows the Krishi Barta flow: a scheduled job collects news from approved sources, the AI cleans, tags and summarizes it, an agri expert verifies it, and it is then published and sent to the farmers whose crop and district match.

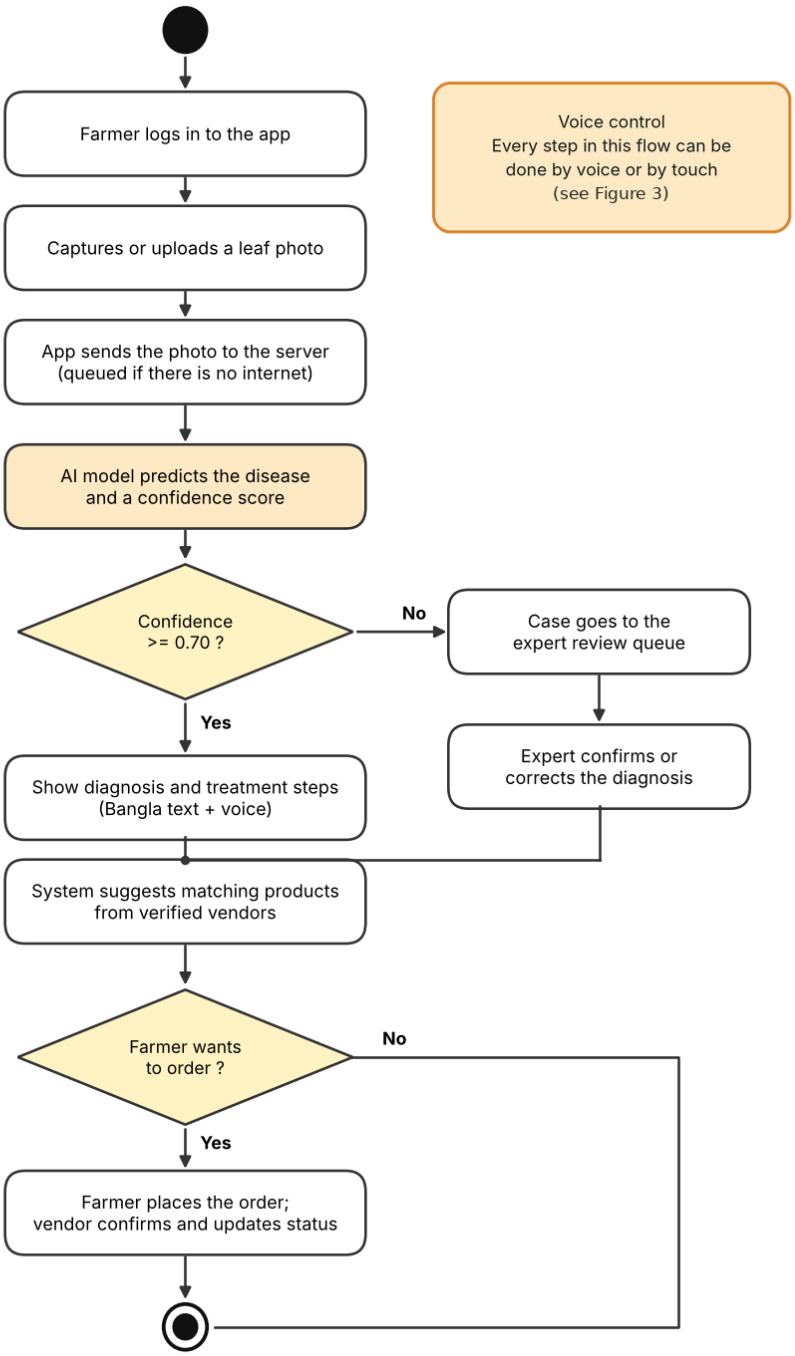


Figure 2: Activity diagram of diagnosis and order flow

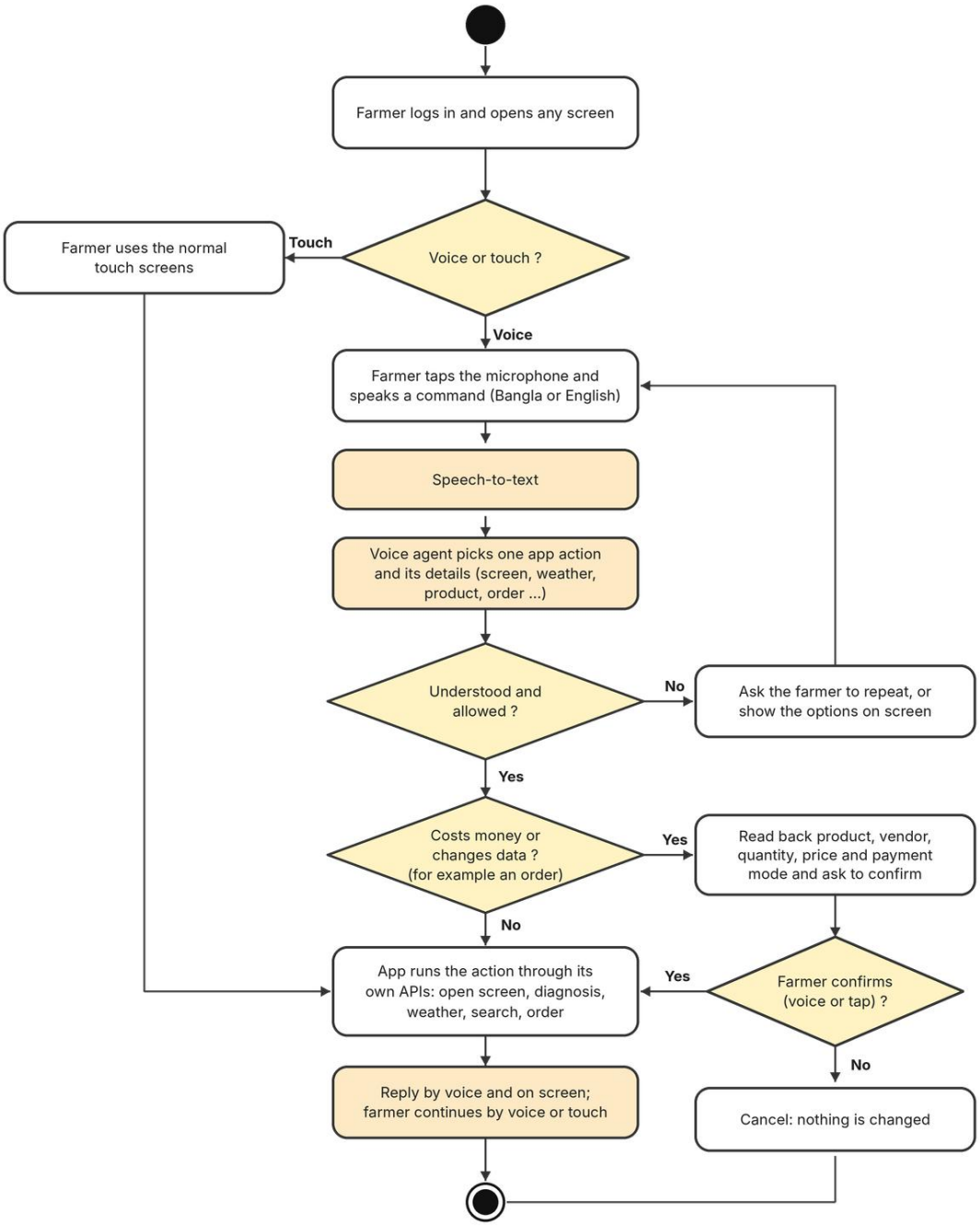


Figure 3: Activity diagram of full voice control of the app

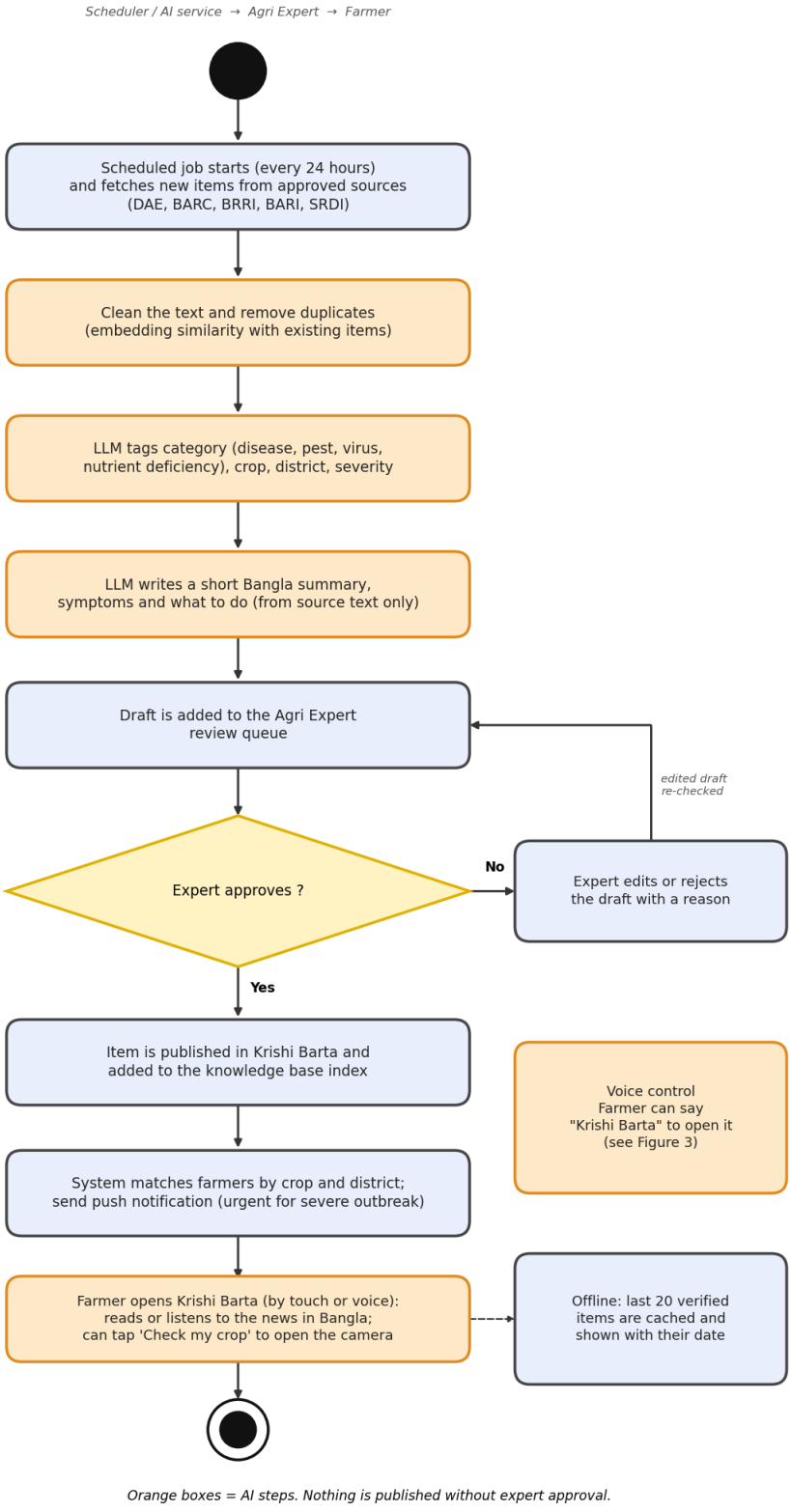


Figure 4: Activity diagram of Krishi Barta news collection, verification and delivery

7.4 Entity Relationship Diagram (ERD)

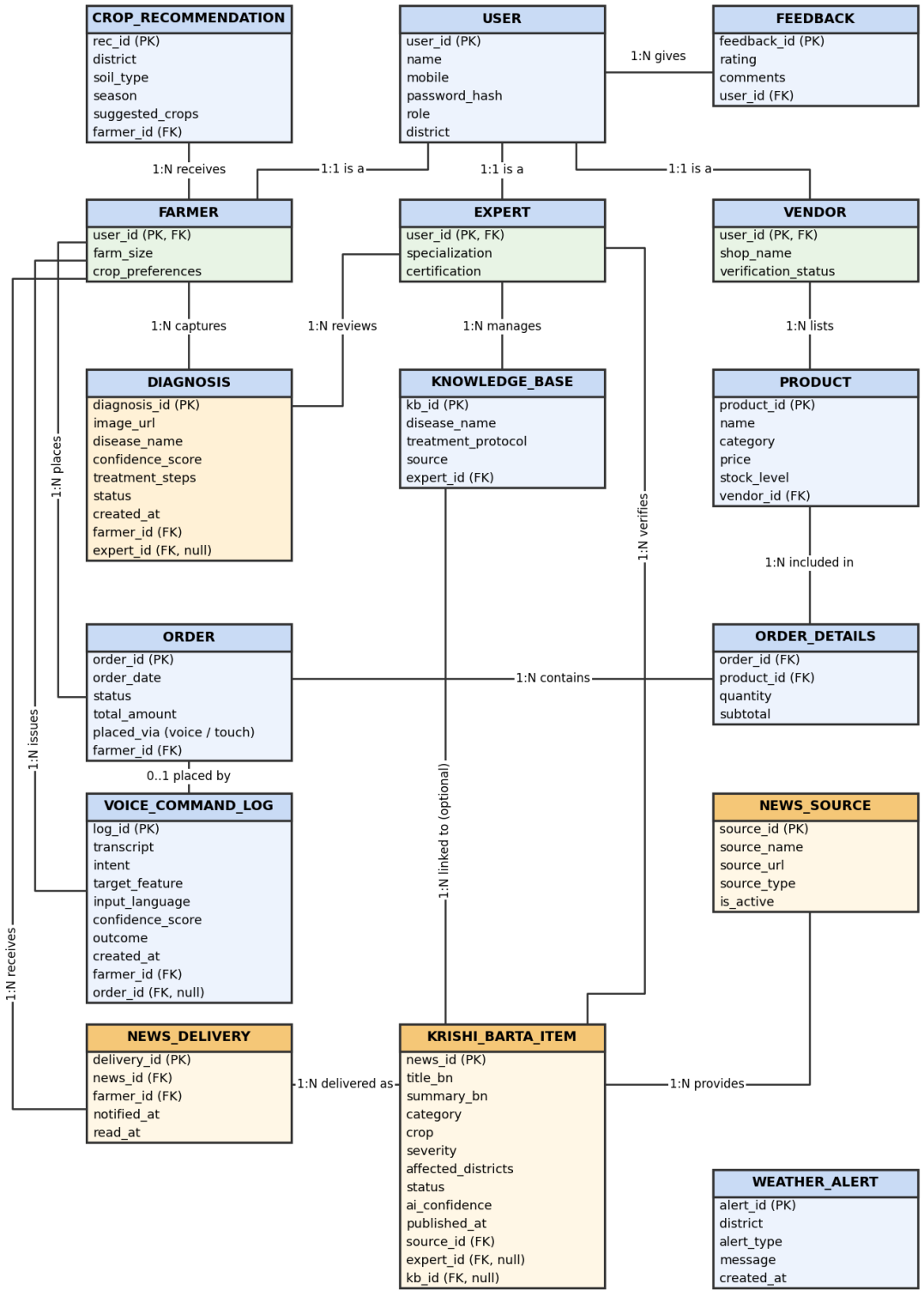


Figure 5: Entity relationship diagram of AgriSense

Entities and attributes

Entity	Attributes
USER	user_id (PK), name, mobile, password_hash, role, district
FARMER	user_id (PK, FK), farm_size, crop_preferences

Entity	Attributes
VENDOR	user_id (PK, FK), shop_name, verification_status
EXPERT	user_id (PK, FK), specialization, certification
DIAGNOSIS	diagnosis_id (PK), image_url, disease_name, confidence_score, treatment_steps, status, created_at, farmer_id (FK), expert_id (FK, optional)
PRODUCT	product_id (PK), name, category, price, stock_level, vendor_id (FK)
ORDER	order_id (PK), order_date, status, total_amount, placed_via (voice or touch), farmer_id (FK)
ORDER_DETAILS	order_id (FK), product_id (FK), quantity, subtotal
KNOWLEDGE_BASE	kb_id (PK), disease_name, treatment_protocol, source, expert_id (FK)
CROP_RECOMMENDATION	rec_id (PK), district, soil_type, season, suggested_crops, farmer_id (FK)
WEATHER_ALERT	alert_id (PK), district, alert_type, message, created_at
FEEDBACK	feedback_id (PK), rating, comments, user_id (FK)
VOICE_COMMAND_LOG	log_id (PK), transcript, intent, target_feature, input_language, confidence_score, outcome, created_at, farmer_id (FK), order_id (FK, optional)
NEWS_SOURCE	source_id (PK), source_name, source_url, source_type, is_active
KRISHI_BARTA_ITEM	news_id (PK), title_bn, summary_bn, category, crop, severity, affected_districts, status, ai_confidence, published_at, source_id (FK), expert_id (FK, optional), kb_id (FK, optional)
NEWS_DELIVERY	delivery_id (PK), news_id (FK), farmer_id (FK), notified_at, read_at

Relationships and cardinality

- USER specializes into FARMER, VENDOR and EXPERT (1:1).
- One FARMER can have many DIAGNOSIS records (1:N).
- One EXPERT can review many DIAGNOSIS records (1:N). A diagnosis has at most one reviewing expert.
- One EXPERT can manage many KNOWLEDGE_BASE entries (1:N).
- One VENDOR can list many PRODUCT records (1:N).
- One FARMER can place many ORDER records (1:N).
- ORDER and PRODUCT have a many-to-many relationship (M:N), resolved through ORDER_DETAILS: one order has many order details (1:N), and one product appears in many order details (1:N).
- One FARMER can receive many CROP_RECOMMENDATION records (1:N).
- One USER can give many FEEDBACK records (1:N).
- One FARMER can issue many VOICE_COMMAND_LOG records (1:N).
- A VOICE_COMMAND_LOG record is linked to at most one ORDER, when the command resulted in an order.

- One NEWS_SOURCE can provide many KRISHI_BARTA_ITEM records (1:N). Each item comes from exactly one source.
- One EXPERT can verify many KRISHI_BARTA_ITEM records (1:N). An item has at most one verifying expert, and none while it is still a draft.
- FARMER and KRISHI_BARTA_ITEM have a many-to-many relationship (M:N), resolved through NEWS_DELIVERY: one farmer has many deliveries (1:N) and one item is delivered many times (1:N). NEWS_DELIVERY stores when the farmer was notified and when the item was read.
- One KNOWLEDGE_BASE entry can be linked to many KRISHI_BARTA_ITEM records (1:N, optional), for example several news items about the same disease.
- WEATHER_ALERT has no foreign key. It is matched to farmers by district.

7.5 AI Feature Design

AI is the core of AgriSense. This section describes the disease detection model in detail, since it drives the main workflow, and then the voice assistant, the voice command agent, the crop recommender and the Krishi Barta news engine.

Data

- **Disease images:** public datasets (PlantVillage for potato and tomato, and a Kaggle rice leaf disease set), plus photos we collect from fields and markets with phone cameras. total image count and local image count when known
- **Labelling:** public images come with labels. Local images are labelled by the team and checked by an agri expert.
- **Preparation:** images are resized to 224 by 224, normalized and augmented (flip, rotation, brightness). The data is split into training, validation and test sets (about 70%, 15% and 15%), keeping the same plant out of more than one set to avoid leakage. Local images are also kept as a separate test set.
- **Knowledge base:** verified DAE, BARC and SRDI documents, split into short passages and stored with embeddings. Each passage keeps its source name.
- **Recommender data:** a district-level table of soil types, seasons and suitable crops, built from SRDI and BARC publications and reviewed by an agri expert.
- **Voice command data:** a list of example Bangla and English commands for each app action (open a screen, weather, product search, order), written with farmers and checked by the team, used to test the agent.
- **Krishi Barta data:** the news items come from official bulletins and advisories of DAE, BARC, BRRI, BARI and SRDI and from items added by the agri expert. An initial set of past items is collected for testing. The agri expert labels each test item with its category (disease, pest, virus or nutrient deficiency), crop and districts. Text is cleaned (markup removed, language checked, duplicates removed) before the AI uses it.

Model

- **Disease detection:** MobileNetV2 with transfer learning. It fits our small dataset and runs fast. The baseline is a small CNN trained from scratch, and we will report how much the transfer learning model improves on it.
- **Voice assistant:** speech-to-text, then RAG with an LLM, then text-to-speech. The baseline is plain keyword search over the same documents.
- **Crop recommender:** random forest plus expert rules. The baseline is a fixed lookup table of crops by district and season.
- **Voice command agent:** speech-to-text, then an LLM with function calling that picks one action from a fixed list and fills in its parameters, then the backend runs the action, then text-to-speech. The baseline is keyword matching of command phrases to actions.
- **Krishi Barta news engine:** multilingual embeddings find duplicate items, and an LLM API tags each item and writes a short Bangla summary from the source text only. This fits because the source bulletins are long and technical while farmers need short, simple Bangla, and an expert can check a short draft quickly. The baseline is keyword rules for the category and crop tags with summaries written by hand, and we will report how much the LLM reduces the expert's editing effort.

Pipeline

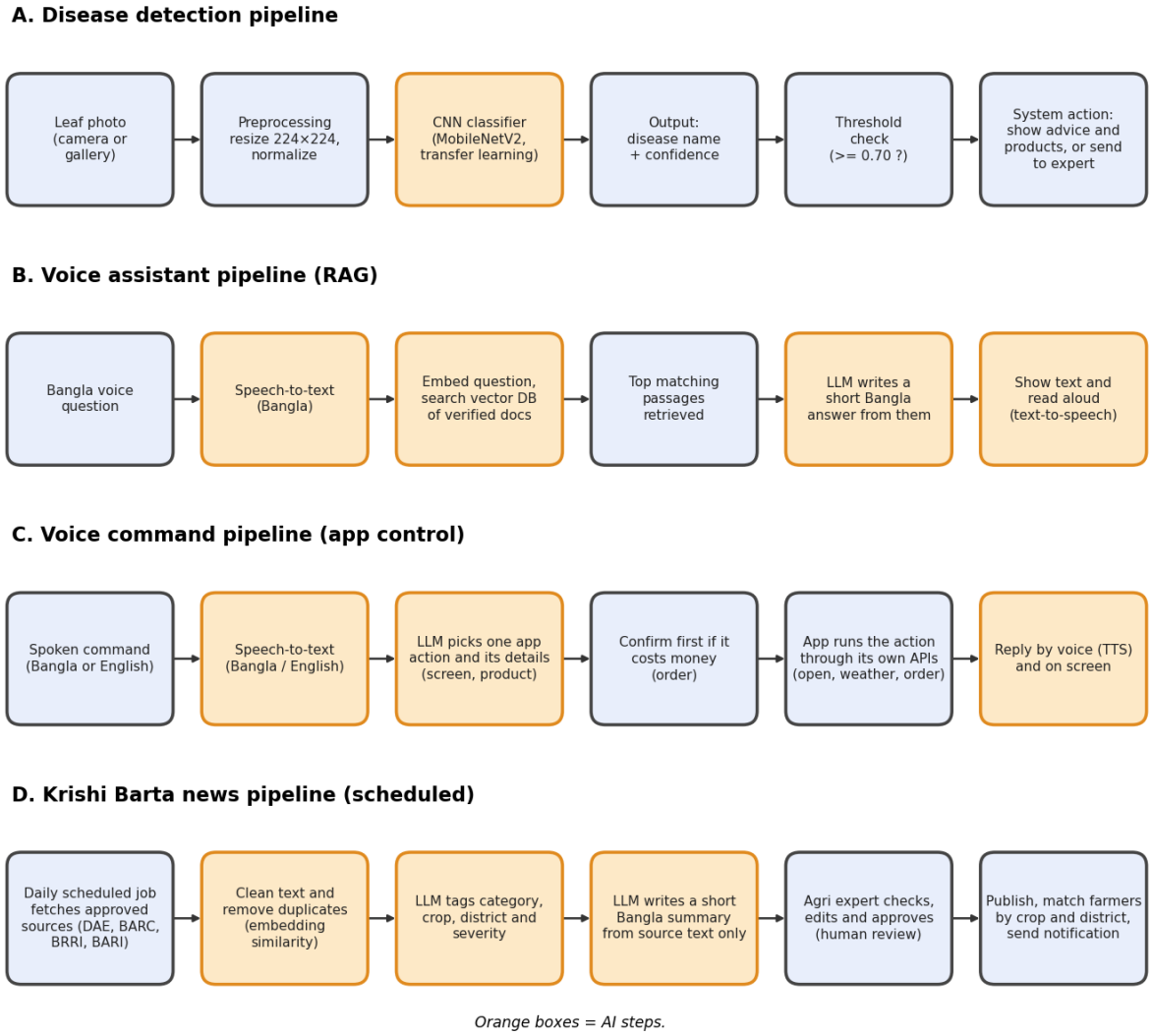


Figure 6: AI pipelines for disease detection, the voice assistant, voice command control and Krishi Barta news

Disease detection: leaf photo, preprocessing, MobileNetV2 prediction, disease name with confidence, threshold check at 0.70, then the system shows advice and products, or creates an expert review case. Crop recommendation: district, soil type, season and weather, then feature preparation, then the random forest ranks crops, then the top three are shown.

Voice command control: spoken command, speech-to-text, intent and parameter extraction, confirmation if the action costs money (placing an order), execution through the existing app APIs (open a screen, fetch weather, search or order a product), then a spoken and written reply.

Krishi Barta news: scheduled job fetches items from the approved sources, then text cleaning and duplicate removal, then the LLM tags category, crop, district and severity, then the LLM writes a short Bangla summary, then the agri expert checks and approves, then the item is published and sent to the farmers whose crop and district match. For example, an approved item with high severity that matches a farmer's crop and district creates a push notification.

Integration

The Flutter app uploads the photo to the Django backend. The backend saves the image, calls the FastAPI model service (/predict) and stores the returned disease name and confidence in

the DIAGNOSIS table. If the score is below the threshold, the backend sets the status to pending review so that it appears in the expert queue. The voice and recommendation requests are handled in the same way, through separate backend endpoints that call the speech services, the retrieval index and the recommender.

Voice commands are sent to a separate backend endpoint. The agent returns a structured action (intent, parameters and confidence). The backend checks it against the farmer's role and runs it by calling the same services that the touch screens use (weather, product search, order creation). An order is created only after the confirmation step. Because voice and touch share these services, the results are identical and the manual route always remains available.

Krishi Barta runs as a background job (for example a Celery task scheduled every day). The job calls the news collector, which fetches the approved sources, and then calls the LLM service for tagging and summary. Each result is saved in KRISHI_BARTA_ITEM with status Draft and the AI confidence, and it appears in the expert's review queue in the web panel. When the expert approves an item, the status changes to Published, the item is added to the retrieval index, and the backend creates NEWS_DELIVERY records and sends push notifications to the matching farmers. The Flutter app reads published items from a /krishi-barta endpoint and keeps the latest 20 on the phone for offline use.

Evaluation and Fallback

- **Disease model:** accuracy on the held-out test set (target 85% or more), per-class precision and recall, a confusion matrix, and a separate result on the locally collected images. Weak classes will be reported openly.
- **Voice assistant:** we will check a set of test questions by hand for correct and grounded answers, and record how often the system correctly says it has no answer.
- **Voice command agent:** intent accuracy on a test set of Bangla and English commands (target 90% or more), correct extraction of product and quantity for order commands, and a check that no order is created without confirmation.
- **Recommender:** top-3 agreement with the agri expert's choices on a set of test cases, compared with the lookup-table baseline.
- **Krishi Barta news engine:** on the expert-labelled test items, accuracy of the category and crop tags (target 90% or more) and the comparison with the keyword baseline, the share of AI summaries that the expert approves without changes, a manual check that no summary contains a fact that is missing from its source, the duplicate detection rate, and the time from collection to publication.
- **When the AI is wrong or unavailable:** low-confidence photos go to an expert and are never shown as final. If the model service is down, the app queues the photo and tells the farmer. If speech recognition fails, the farmer can type. If retrieval finds nothing, the assistant says so. If the weather API fails, the recommender uses district and season only. If the voice command agent cannot decide the intent, it asks the farmer to repeat or shows the options on screen. If the voice services are down, the farmer continues with the touch screens. If a news source cannot be reached, the job retries later and the app keeps showing the last verified items with their date. If the LLM is unavailable or

not confident, the raw item goes to the expert, who can write and publish it by hand. An item that the expert has not approved is never published.

8. Team Member Responsibility Matrix

Member Name	ID	Primary Responsibilities
Abdullah-Al-Arafat Sirat	241-35-057	Frontend: Flutter app and Bangla UI/UX, voice input and output screens, marketplace screens, high-contrast mode, state management and offline photo queue. Secondary: usability testing and farmer feedback. Voice command button, confirmation screens and voice and manual mode switching. Krishi Barta screens, notification settings and the offline news cache.
Aranna Saha	241-35-197	AI: image collection and preprocessing, disease model training and evaluation, crop recommender, RAG pipeline and knowledge base, model service. Secondary: helping with model deployment on the backend. Voice command agent (intent recognition and test set). Krishi Barta news tagging and summarization pipeline and its test set.
Sabbir Hasan Emon	241-35-012	Backend: Django REST APIs, database schema, authentication and security, integration of AI services, project planning. Secondary: documentation and final report. Voice command endpoint, action validation and command log. Krishi Barta scheduler, source collector, expert review API and notification delivery.

All members share system testing, field testing, requirement validation with experts and presentation preparation.

Appendix A: Elicitation Evidence

https://drive.google.com/drive/folders/1KO_zgWEondKulr32ebyA1C1Eou8q5YsB?usp=drive_link